classical E-L equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$\delta \mathcal{L}(x) = \mathcal{L}(\delta x)$.

$\delta \mathcal{L}(\dot{x}) = \mathcal{L}(\delta \dot{x})$

No V only $T = \frac{1}{2} m \dot{x}^2$

$\mathcal{L} = \frac{1}{2} m \dot{x}^2$

what if we have a new

$\bar{\mathcal{L}} = \frac{1}{2} m \dot{x}^2 + \dot{x} g(x)$

$$\frac{\partial \bar{\mathcal{L}}}{\partial \dot{x}} = m\dot{x} + g(x)$$

$$\frac{d}{dt}\frac{\partial \bar{\mathcal{L}}}{\partial \dot{x}} = m\ddot{x} + g(x)\dot{x} \quad \text{(chain rule)}$$

$$\frac{\partial \bar{\mathcal{L}}}{\partial x} = g(x)\dot{x} \quad \text{(chain rule)}$$

$$\frac{d}{dt}\frac{\partial \bar{\mathcal{L}}}{\partial \dot{x}} - \frac{\partial \bar{\mathcal{L}}}{\partial x} = m\ddot{x}, \text{ same!}$$

How about another $\bar{L}$?

$$L \qquad \bar{L} = L + \frac{dt}{dt}$$

$$I = \int_{t_1}^{t_2} L\,dt \qquad \bar{I} = \int_{t_1}^{t_2} \left(L + \frac{dt}{dt}\right) dt$$

$$= \int_{t_1}^{t_2} L\,dt + \int_{t_1}^{t_2}$$

$$I = \int L\,dt \qquad \boxed{\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0 \rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}}$$

$t' = t + \delta t$

$L' = L + \frac{dL}{dt}\delta t$

$\frac{dL}{dt} = \frac{\partial L}{\partial x}\frac{dx}{dt} + \frac{\partial L}{\partial \dot{x}}\frac{d\dot{x}}{dt}$ chain rule

$= \frac{d}{dt}\frac{\partial L}{\partial \dot{x}}\frac{dx}{dt} + \frac{\partial L}{\partial \dot{x}}\frac{d\dot{x}}{dt}$

chain rule again:

$\frac{d}{dt}(fg) = \frac{df}{dt}g + f\frac{dg}{dt}$

$f = \frac{\partial L}{\partial \dot{x}}$

$g = \dot{x}$

$$\frac{dL}{dt} = \frac{\partial L}{\partial x}\frac{dx}{dt} + \frac{\partial L}{\partial \dot{x}}\frac{d\dot{x}}{dt} \quad \text{(chain rule)}$$

$$= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\frac{dx}{dt} + \frac{\partial L}{\partial x}\frac{d\dot{x}}{dt}$$

$$= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\dot{x} + \frac{\partial L}{\partial \dot{x}}\frac{d\dot{x}}{dt}$$

↓ ~~chain rule~~ product rule

$$= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\dot{x}\right)$$

$$\frac{d}{dt}(fg) = \frac{df}{dt}g + \frac{dg}{dt}f$$

$$f = \frac{\partial L}{\partial \dot{x}} \qquad g = \dot{x}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\dot{x}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\dot{x} + \frac{\partial L}{\partial \dot{x}}\ddot{x}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\dot{x}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\dot{x} + \frac{\partial L}{\partial \dot{x}}\frac{d\dot{x}}{dt}$$

that's

$$\frac{dL}{dt} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\dot{x}\right) = 0$$

$$\frac{d}{dt}\left(L - \frac{\partial L}{\partial \dot{x}}\dot{x}\right) = 0$$

or

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\dot{x} - L\right) = 0$$

$$\frac{\partial L}{\partial \dot{x}}\dot{x} - L = C$$

what's that?

$$L = T - U = \frac{1}{2}m\dot{x}^2 - mgh$$

$$\frac{\partial L}{\partial \dot{x}}\dot{x} = m\dot{x}^2 \qquad \frac{\partial L}{\partial \dot{x}}\dot{x} - L = \underbrace{\frac{1}{2}m\dot{x}^2 + mgh}_{\text{total } E}$$

Emmy Noether

Generalized Symmetry



"mirror"



"rotation"

$f(x) = x^4 + 3x + 2$

offset

$= x^4 + 3x + 2000$

Action of "mirror"

Action of rotation

$\frac{df}{dx}$ stays the same

Action of translation symmetry.

what L's invariance tells us:

$\frac{\partial L}{\partial x} = 0 \quad \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = C$ translation symmetry ↪ conservation of momentum

$\frac{\partial L}{\partial \theta} = 0 \quad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = C$ rotation symmetry ↪ conservation of angular momentum

Time translation symmetry ↪ conservation of energy